

Sleep, Screen Time, and BMI Percentile

YRBS 2007 National High School Survey | ANOVA + weighted regression

Analysis Sample

11,256

complete student records

Short sleep

+2.36

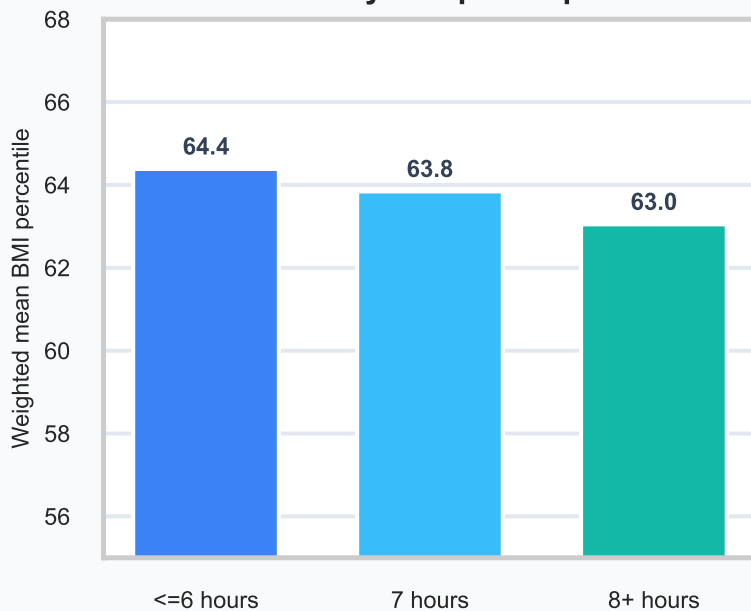
BMI percentile vs 8+ hours

TV time

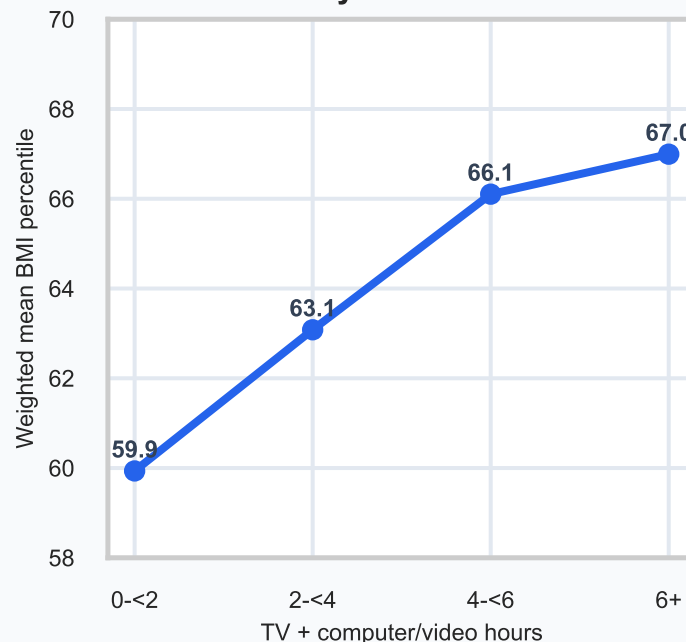
+2.16

BMI percentile per hour

BMI by Sleep Group



BMI by Screen Time



Research Question

Are sleep duration and lifestyle behaviors associated with BMI percentile?

Outcome: BMI percentile

Predictors: sleep, TV, computer/video use, physical activity, sex, and grade

Adjusted Regression Results

+2.36 Short sleep vs 8+ hours | $p=0.005$

+2.16 Each additional TV hour | $p<0.001$

0.021 Model R-squared

Takeaway

Shorter sleep and higher TV time are associated with higher BMI percentile. This is an association in a cross-sectional survey, not proof of causation.

ANOVA p-value for sleep groups: 0.029. Weighted regression uses survey weights and robust HC3 standard errors.